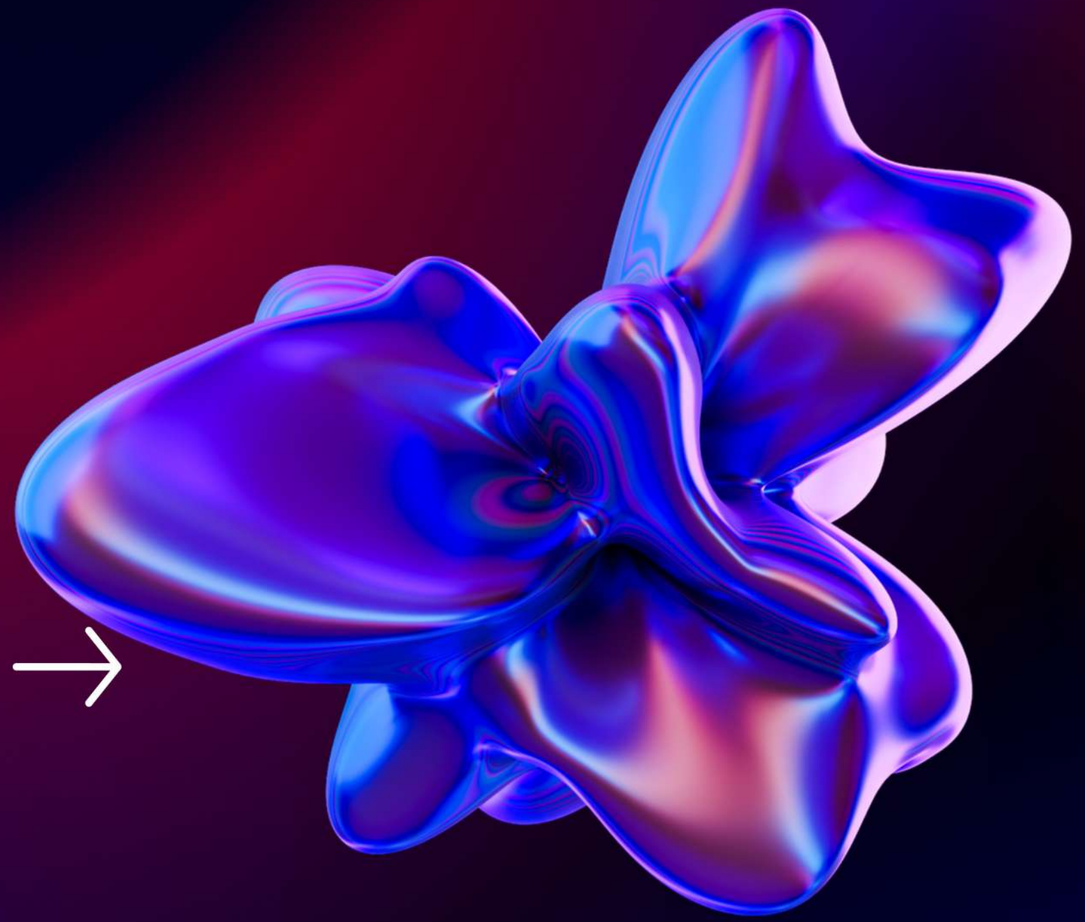
CODESTOME

HACKATHON 2K26



Prepared by: Shivam
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Meet
Dixa

LIBRARIES

01 | Main Language “ Python ”

02 | EasyOCR

03 | Streamlit

04 | OpenCV

05 | NumPy

06 | Pillow

07 | Conclusion

PROJECT OBJECTIVES

Goals of the Project

The objective of this project is to develop an intelligent AI-based Optical Character Extraction (OCE) system capable of accurately detecting, extracting, and processing textual information from various sources such as images, documents, and scanned files.

The system aims to achieve high accuracy in text recognition while supporting maximum language conversion capabilities, enabling seamless translation and interpretation across multiple languages. Additionally, it focuses on detailed text identification, ensuring precise extraction of structured and unstructured data for further analysis.

- **To gain an in-depth understanding of the AI-based OCE system and its applications**
- **To enhance teamwork and coordination among project members**
- **To successfully develop and deliver an efficient and accurate system**

PROJECT METHODOLOGY

USE OF AI

Topic selection and planning

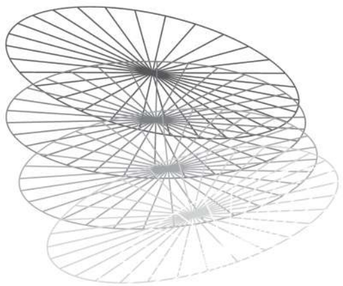
The project begins with identifying the scope and objectives of the AI-based OCE system. Proper planning is carried out to define system requirements, tools, and technologies needed for accurate text extraction and processing.

Research and information gathering

Extensive research is conducted on AI techniques such as Optical Character Recognition (OCR), Natural Language Processing (NLP), and language conversion methods. Relevant data and resources are collected to build an efficient and accurate system.

CHALLENGES & SOLUTIONS

Problems Faced During the Project



Time management issues

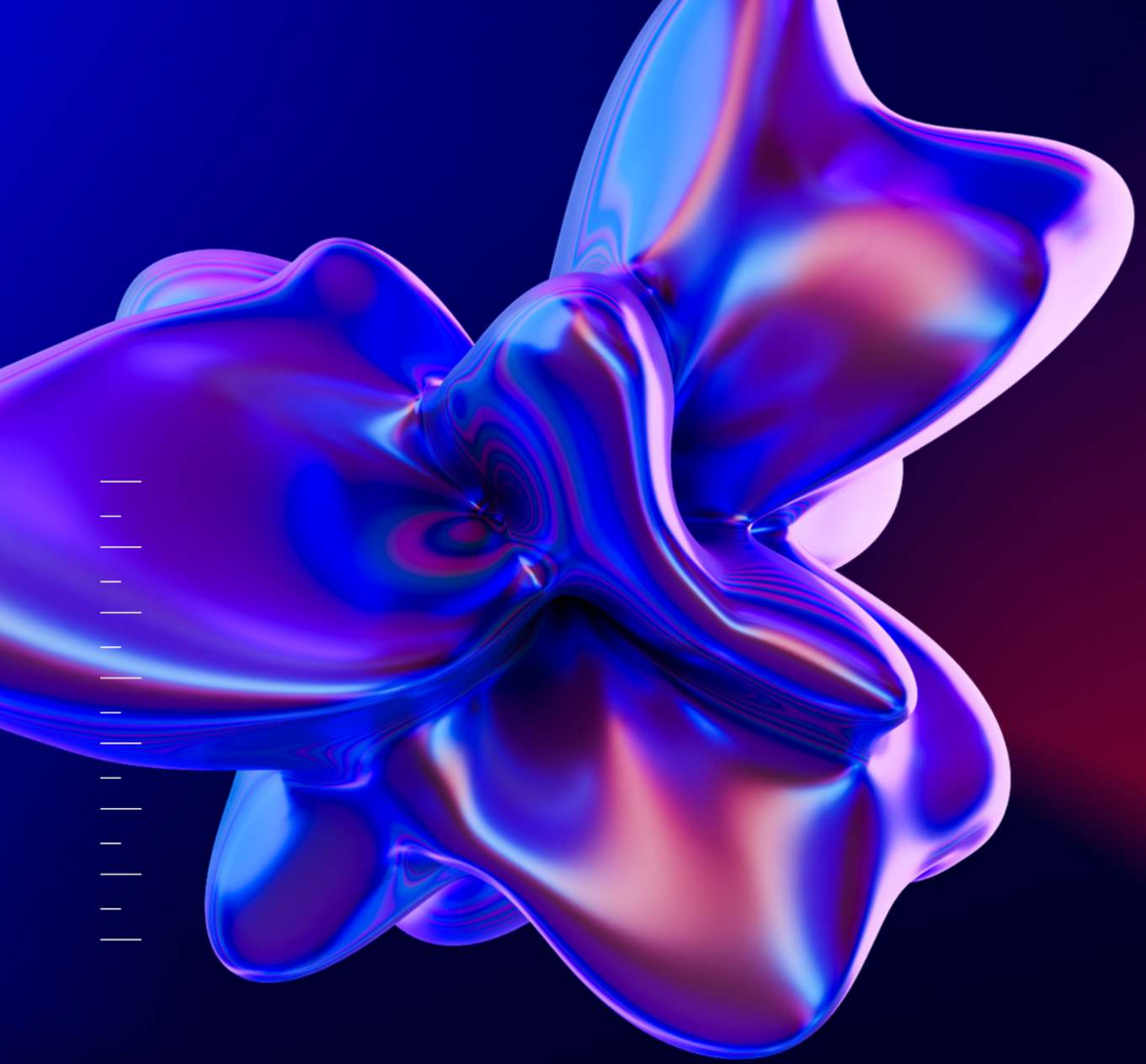
Managing time effectively was a challenge due to multiple project phases such as data collection, model training, and testing.
Solution: A proper schedule and task distribution were implemented to ensure timely completion of each phase.

Communication gaps

Lack of clear communication among team members sometimes led to misunderstandings and delays in progress.
Solution: Regular meetings and clear task assignments improved coordination and teamwork.

Technical difficulties

Challenges were faced in implementing AI models, handling different image formats, and ensuring accurate text extraction and language conversion.
Solution: Continuous research, testing, and use of suitable tools and libraries helped overcome technical issues and improve system performance.



THANK YOU

Questions & Discussion

Thank you for your time and attention